PENGUIN ICE RUN.

Game Design Document · Assignment 8–9
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Unity 6.6 · github.com/AkamenYY/penguin-ice-run

A little penguin runs and jumps across floating ice floes to win back the fish that the Walrus King stole.

1. TITLE + PITCH

Penguin Ice Run — a short, bright arctic adventure: collect fish, open the portal, beat the Walrus King.

2. GENRE · PLATFORM

3D action platformer. **Android** (iOS later), **landscape**. Short sessions: one level ≈ 2–4 minutes.

3. 2D OR 3D

3D, third-person camera behind the hero, movement relative to the camera. The hero is a penguin with an Animator (Idle, Run, Jump, Wave). In week 10 the current low-poly model is replaced by a **rigged penguin model** using the same Animator states.

4. CORE LOOP

RUN → **JUMP** → **COLLECT** → **THROW** ↺

Run across the floes, jump over the water, collect fish, throw snowballs at enemies. Enough fish open the portal → next level. The last level is the boss.

5. CONTROLS (ON-SCREEN)

Button	Where	What it does
Joystick	bottom-left	move (relative to camera)
JUMP	bottom-right	jump
FIRE	bottom-right	throw a snowball forward (short cooldown)
II (Pause)	top-right	pause: Resume / Restart / Menu

Keyboard in parallel for testing: WASD, Space, X.

6. CONTENT

	Now (midterm build)	Target for week 15
Levels	3: tutorial, seals + icicles, boss arena	4 levels + boss arena, level select
Enemies	patrolling seal, falling icicles, Walrus King boss (2 phases)	+ chasing seal with AI states (patrol → chase → return)
Items	fish (goal), heart (medkit)	+ golden fish (bonus score)
Hazards	ice spikes, water (fall = restart), moving floes	+ cracking floe (falls after 1 s)

7. ART AND SOUND

Low-poly, flat colours, one arctic world: penguin, seals, walrus, fish and ice are made in Unity from primitives and simple procedural meshes (own work). Procedural sky: day → sunset → night for the boss. UI: bold white text with outline, big rounded orange/blue buttons — same style on every screen.

Sound: jump, pickup, hurt, throw, boss attack, win + looping music (quieter than effects). Week 12: extend with Kenney Audio (CC0) and add volume settings.

8. MVP — MUST BE IN THE WEEK-15 BUILD

- ✓ Main menu, HUD, pause — on 16:9, notch phones and tablets
- ✓ 4 levels + boss, death → restart, win screen
- ✓ Enemy AI (seal states), boss with 2 phases
- ✓ Saving: unlocked levels, best fish, settings
- ✓ Sound + music with volume settings
- ✓ Particle effects; 30+ FPS on an Android phone

9. CUT LIST — I WILL NOT MAKE

- ✗ Online multiplayer or leaderboards
- ✗ Shop, coins, in-app purchases
- ✗ Skins / character customisation
- ✗ Procedurally generated levels
- ✗ Story cutscenes and dialogues
- ✗ More than one boss
- ✗ App Store / Google Play release

10. PLAN, WEEKS 9–13

Week	Ready at the end of the week
9	GDD, GitHub repo, mobile UI: menu, HUD, pause, SafeArea, 3-device check
10	Enemy AI: seal patrol → chase → return (NavMesh); rigged penguin model
11	Saving (PlayerPrefs / JSON): unlocked levels, best fish; level select
12	Sound: AudioMixer, music & SFX sliders, new effects
13	Effects & polish: particles, camera shake, Level 4

Roles: solo until week 14 — Madiyar Tokbergen: design, programming, art, sound. Teams form in week 14.